

Agreement between Phenotypically Detected Linezolid Resistance and Mutations in *rrl* and *rplC* Genes of *Mycobacterium tuberculosis* Isolates Using Nanopore Sequencing

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Abstract

Background: Phenotypic drug susceptibility testing (DST) is considered the gold standard for detecting linezolid (LZD) resistance in *Mycobacterium tuberculosis* (MTB), but it is time-consuming. Nanopore sequencing offers a potentially faster alternative approach. This study evaluated the agreement between phenotypically detected LZD resistance and mutations in the *rrl* and *rplC* genes of MTB isolates using nanopore sequencing. **Methods:** Consecutive drug-resistant MTB isolates from pulmonary samples collected in 2021 underwent liquid culture (LC) DST for LZD. All resistant isolates and an equal number of susceptible isolates were subjected to targeted sequencing of the *rrl* and *rplC* genes using nanopore technology. **Results:** Sequencing identified a C154R mutation in the *rplC* gene in only one LZD-resistant isolate. No mutations were detected in the *rrl* gene. The agreement between sequencing and LC-DST for detecting LZD resistance was poor (Cohen's kappa: 0.03571, 95% confidence interval [CI]: -0.034–0.105). Additionally, no significant association was found between LZD resistance and clinical or microbiological outcomes at 6-month follow-up. **Conclusion:** This study revealed a considerable discrepancy between phenotypic and genotypic detection of LZD resistance in MTB. Further research is needed to better understand the genetic mechanisms underlying LZD resistance and to develop reliable molecular diagnostics for rapid resistance detection.

Keywords: Gene mutation, linezolid, multidrug-resistant tuberculosis, *Mycobacterium tuberculosis*, nanopore sequencing

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INTRODUCTION

Linezolid (LZD), the first oxazolidinone approved for clinical use, was originally used to treat infections caused by Gram-positive bacteria.^[1] Later, it was also found to improve the treatment success rate of multidrug-resistant (MDR) tuberculosis (TB) threefold and was therefore approved for its treatment in 2016 by the World Health Organization (WHO).^[2]

LZD has undergone several clinical trials in combination with bedaquiline (B) and pretomanid (Pa) Bedaquiline, Pretomanid, Linezolid (BPAL) regimen for the treatment of drug-intolerant or nonresponsive MDR/preextensively drug-resistant TB (pre-XDR TB). The first trial of BPAL regimen was the Nix-TB phase 3 clinical trial that was conducted in 107 patients with highly drug-resistant TB from South Africa. This was followed by the ZeNix trial, in which 181 participants from South

Africa, Georgia, Russia, and Moldova were enrolled.^[3] However, the long-term use of the protein synthesis inhibitor LZD was linked to negative side effects in this regimen; therefore, another protein synthesis inhibitor of *Mycobacterium tuberculosis* (MTB) spectinamide 1599 was added to BPAL regimen as it had lesser side effects (bedaquiline Pa spectinamide regimen).^[4] In India, a modified BPAL trial is being conducted to study the safety of varying doses of LZD with bedaquiline and Pa in adults with

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preextensively drug-resistant (XDR) or treatment-intolerant MDR pulmonary TB.^[5] The results of this trial are awaited.

Despite its recent introduction in the treatment regimen of MDR TB, resistance to LZD has already been reported from different parts of the world. A recent meta-analysis showed the pooled frequency of LZD resistance to be 4.2% among the clinical isolates of MDR MTB from studies across Asia, Europe, and the United States of America.^[6] In South Africa, among the patients showing failure to LZD-based treatment regimens, LZD resistance was detected in 13 of 39 patients.^[2] A study from Mumbai, India obtained LZD resistance in 68 of 2179 isolates.^[6]

LZD acts by preventing the formation of 70S initiation complex in the mycobacterial cell by binding to the 50S ribosomal subunit near its interface with 30S subunit, thereby inhibiting bacterial protein synthesis.^[6] Mutations in ribosomal genes that produce 50S ribosomal subunit, chiefly the *rplC* gene encoding L3 ribosomal protein or the *rrl* gene encoding 23S rRNA, have been principally found to confer LZD resistance. These mutations lead to conformational changes in the rRNA at LZD binding sites, thereby rendering it unable to bind to the peptidyl transferase center.^[6] Other mechanisms such as overexpression of efflux pumps and decreased permeability of mycobacterial cell wall because of *Fad32* gene mutations have also been reported to contribute to LZD resistance in mycobacteria.^[6]

Currently, for detection of LZD resistance, automated phenotypic culture-based method using the BACTEC MGIT 960 system at a critical concentration of 1 µg/mL is considered as the gold standard. Therefore, the guidelines for programmatic management of drug resistant tuberculosis (PMDT) recommend conducting phenotypic LC-DST for detecting LZD susceptibility in cases where resistance to rifampicin and/ or isoniazid is detected by nucleic acid amplification tests or LPA. However, the major limitation of LC-DST is that susceptibility results are available in weeks to months that delay appropriate treatment initiation.^[7] Early detection of LZD resistance would help clinicians in designing patient-tailored regimen according to the drug replacement sequence prescribed by the WHO.^[2] Besides giving delayed results, LC-DST also have a disadvantage of not being able to detect clinically significant resistance in certain isolates as has been observed for the first-line anti-tubercular drugs. In such cases, molecular detection of drug resistance seems to be better for predicting patient outcomes.^[8] This phenomenon may occur with LZD LC-DST also, but it is yet not known. The currently available WHO-approved commercial molecular methods including GeneXpert, TrueNat, and line probe assays are not standardized for detecting LZD resistance. Hence, whole-genome sequencing (WGS) seems to be the only possible option for obtaining rapid and reliable results for LZD resistance. WGS by Illumina platform is being used worldwide for detecting resistance in MTB, but its disadvantages are high instrument cost, sophisticated settings for installation, and errors in reading larger fragments. Nanopore sequencing technology overcomes some of these limitations. Besides being portable and easy to set up even in

resource-limited settings, nanopore sequencing offers an advantage of long sequence reads and real-time data analysis, due to which reactions can be stopped when sufficient data have been gathered. Nanopore sequencing has been tried for MTB for detecting drug resistance in several countries with variable results,^[9] but data on its effectiveness in the Indian context is very sparse.^[9]

Looking at the advantages of nanopore sequencing, the authors hypothesized that it can be used as an alternative to phenotypic DST (pDST) for LZD in the clinical settings. This study was therefore done to determine the frequency of primary LZD resistance at a tertiary care center in North India and to estimate the agreement between phenotypically detected LZD resistance and mutations in *rrl* and *rplC* genes of MTB isolates using nanopore sequencing. The aim was also to study clinical and microbiological outcomes in patients with LZD-resistant TB.

METHODS

Ethical consideration

This study was approved by the Institutional Ethics Committee (2111/Ethics/2023) on February 27, 2023. Written informed consent was taken from all the participants.

Type of sampling and reasons for selection

It was an observational study, which was conducted at the Intermediate Reference Laboratory for TB, Department of Microbiology, King George's Medical University, Lucknow, Uttar Pradesh in the year 2023. The laboratory performs quality work and is accredited by the National Accreditation Board for Testing and Calibrating Laboratories (NABL) and by the National TB Elimination Program (NTEP), India.

A total of 527 well characterized, stored MTB isolates that were obtained sequentially from pulmonary samples in the year 2021 were selected. The preserved culture isolates were revived by subculturing on Lowenstein–Jensen slopes following the WHO Mycobacteriology Laboratory manual 2014 and incubating for 4 weeks or until confluent growth of MTB was obtained.^[10] Mycobacterial growth indicator tubes (MGITs) were inoculated from the growth obtained, which were then subjected to repeat pDST. The selected isolates were then subjected to genotypic drug resistance testing for LZD.

Patient consent statement

Written informed consent was taken from all the patients.

Inclusion criteria

Consecutively obtained isolates of MTB from patients fulfilling the following criteria were included in the study:

1. Patients with pulmonary TB who had never received LZD
2. Resistant to first- and/or second-line drugs, as determined by GenoType MTBDR plus assay and GenoType MTBDRsl assay (Hain Lifescience, Nehren, Germany), respectively
3. For whom clinical outcome data was available at 6-month follow-up
4. All LZD-resistant isolates ($n=28$) and an equal number

of consecutively obtained susceptible isolates ($n=28$) as determined by pDST (for whom patient clinical outcome data was available at 6-month follow-up) during the same time period.

Exclusion criteria

Following MTB isolates were excluded from the study:

1. Isolates that had a low bacterial load (faint turbidity on MGIT-960 tube)
2. *M. tuberculosis* positive cultures along with contamination

Phenotypic drug susceptibility testing

It was done by BACTEC MGIT 960 system (Becton Dickinson, USA) as per the diagnostic algorithm given by the NTEP 2021 guidelines^[11] and following the techniques mentioned by the WHO.^[10] LZD susceptibility was estimated using critical concentration of 1.0 µg/mL. Isolates were used at an inoculum corresponding to 1 McFarland standard.

Nanopore sequencing

Pure cultures of the selected isolates were subjected to cetyltrimethylammonium bromide protocol for DNA extraction, and extracted DNA was quantified using Nanodrop 8000 (Thermo Fisher Scientific, USA) as per the protocol described by Gliddon *et al.*^[12] Previously published primers targeted against *rplC* and *rrl* genes were used. For *rplC*, the primers used were forward primer 5'ACACAAGCG GTTGATCGAC3' and reverse primer 5'CGTCTTGACGT CGATTTTGAGT3'. The primers targeted against *rrl* gene were forward primer 5'TTGTAAGTGTC TAAGGGCGCATG3' and reverse primer 5'TGTTGTAAGTT TCGGCCGGTTAGCA3'.^[13]

Nanopore sequencing was performed using the Rapid Barcoding Kit 96 V14 (SQK-RBK114.96) following the manufacturer's instructions on the MinION sequencer (Oxford Nanopore Technologies) platform with flow cell (R9.4.1) FLO-MIN106D. The sequences obtained were analyzed using TB-profiler pipeline,^[12] which allows analysis of WGS data of MTB. This interpretation pipeline includes resistance mutations from the currently followed second edition of the catalog of mutations in MTB complex (MTBC) and their association with drug resistance (<https://iris.who.int/handle/10665/374061>).

The treatment regimen of the patients from whom these isolates were obtained was studied. In India as per the PMDT guidelines,^[14] two regimens are used for the treatment of M/XDR TB including shorter oral bedaquiline-containing MDR/rifampicin resistant (RR)-TB regimen consisting of the 4–6 months of the intensive phase of bedaquiline, levofloxacin, clofazimine, pyrazinamide, ethambutol, high-level isoniazid, and ethionamide followed by 5 months of continuation phase of levofloxacin, clofazimine, pyrazinamide, and ethambutol. The second regimen is the longer oral M/XDR-TB regimen consisting of 18–20 months of levofloxacin, LZD, clofazimine, and cycloserine along with 4 months of bedaquiline extendable up to 6 months. The patients' interim clinical and microbiological outcome data at 6-month follow-up was

analyzed to study any association between the patient outcome and LZD resistance.^[14]

Microbiological interim outcomes were defined as bacteriological conversion or culture positive. A patient was labeled as bacteriologically converted when at least two consecutive cultures collected 6 months after initial confirmation of MTB and taken on different occasions at least 7 days apart were found to be negative.^[14] In contrast, if either or both of these consecutive cultures were found to be positive, the patient was labeled as culture positive.

Clinical outcome was defined in terms of died and cured. As per the PMDT guidelines, cured was defined as a pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who completed treatment as recommended by the national policy and has evidence of bacteriological response and no evidence of treatment failure. Died was defined as a patient who died before starting or during treatment.^[14]

Statistical analysis

SPSS version 24.0 (IBM SPSS statistics Chicago, Illinois, United States) was used for statistical analysis. The values were calculated as proportions. Odds ratio was calculated to know the association between the outcome of patients and LZD resistance, and the Chi-square test was calculated to estimate the significance of the regimen with LZD resistance. Agreement between tests for detecting LZD resistance, i.e., nanopore sequencing and LC-DST, was calculated using Cohen's kappa statistic. Sensitivity and specificity were also calculated.

Odds ratio was used to study the strength of the association between risk factors and outcomes in the study. The *P* value was calculated for the hypothesis testing in the study.

The Chi-square was calculated to compare observed results with the expected results.

RESULTS

Baseline LZD DST data were available for 527 MTB isolates obtained from different patients of which 31 (5.9%) were mono-resistant to rifampicin, 90 (17.1%) were mono-resistant to isoniazid, 154 (29.2%) were MDR, and 252 (47.8%) were pre-XDR. Of these 527 isolates, 28 (5.31%) were resistant to LZD by pDST using MGIT960, which on revival and repeat DST testing gave the same results. The distribution of LZD resistance with respect to the drug resistance category of MTB isolates is shown in Figure 1. These 28 resistant isolates were obtained from 18 (64.28%) males and 10 (35.71%) females with age ranging from 9 to 66 years, of which 3 (10.71%) were less than 15 years of age. The 28 LZD susceptible isolates were obtained from 21 (75%) males and 7 (25%) females ranging from 16 to 59 years of age.

Nanopore sequencing detected signature C154R (cysteine 154 arginine) mutation in the *rplC* gene (50S ribosomal protein L3) in only 1 (3.57%) of the 28 LZD-resistant isolates and in

none of the susceptible isolates. Mutation in the *rrl* gene (23S rRNA) was not found in any isolate. These isolates with no mutations in *rplC* and *rrl* genes are referred to as “wild type” in the supplementary file. Agreement between nanopore sequencing and LC-DST for detecting LZD resistance was found to be none to slight (Cohen’s kappa value: 0.03571, standard deviation: 0.03537, 95% confidence interval [CI]: -0.034–0.105). Positive agreement between nanopore sequencing and LC-DST for detecting LZD resistance was 3.57% (1/28) and negative agreement was 100% (28/28). The sensitivity between nanopore sequencing and LC-DST for detecting LZD resistance was 3.57 (95% CI = 0.09%–18.35%) and specificity was 100% (95% CI = 0.09%–18.35%).

Interim patient outcome was obtained at 6 months of the baseline sample for patients with LZD-susceptible and LZD-resistant TB. In both groups, 10 had expired and 18 had survived. Follow-up cultures could be obtained for 12 LZD-resistant and 14 LZD-susceptible patients, of which 9 were culture positive and 17 were culture converted respectively [Table 1]. It was observed that mortality (Odds ratio = 1.00, 95% CI = 0.34–2.98) or culture positivity (Odds ratio = 3.6667, 95% CI = 0.6659–20.1911) was not significantly associated with LZD resistance.

A shorter oral regimen was initially given to 26 out of 28 patients with LZD-resistant MTB. This regimen was also given to 27 out of 28 patients with LZD-susceptible MTB. But during the course of treatment, the regimen change from shorter oral regimen to oral longer regimen was statistically higher ($P = 0.0383$) in patients with LZD resistance as compared to those with LZD-susceptible infection ($\chi^2 = 6.52$, $P = 0.04$) [Table 1]. The association of LZD resistance with the clinical and microbiological outcome of patients at 6-month follow-up and with the treatment regimen is shown in Table 1.

The demographic details, drug resistance pattern, sequencing results, treatment regimen, treatment duration, and clinical and microbiological patient outcomes of the LZD-resistant patients are given in Supplementary Table 1.

DISCUSSION

LZD is being observed as a milestone in the treatment of drug-resistant TB. However, baseline resistance to LZD has been reported from different parts of the world. A recent meta-analysis showed that there is considerable geographic variation in LZD resistance ranging from 22.2% in Spain to 0.2% in the United States, with overall frequency estimated to be 4.2% among clinical isolates of MDR-TB.^[15] The frequency of LZD resistance found in the present study (5.3%) corroborates well with that reported from Mumbai (3.1%, 68 of 2179 isolates).^[6] Stored isolates of the year 2021 were selected for this study since we aimed to determine the baseline LZD resistance. Patients enrolled in the year 2021 were confirmed not to have had any exposure to LZD because at that time the drug was not incorporated in NTEP. The fact whether these patients were exposed to LZD due to some other bacterial infections is not known.

Nanopore sequencing identified C154R mutation in the *rplC* gene (50S ribosomal protein L3) in only one isolate and no mutation in the *rrl* gene (23S rRNA). A previous study from Mumbai showed *rplC* gene and *rrl* gene mutations in 16 and 3 LZD-resistant isolates, respectively, of the 32 isolates sequenced.^[6] Furthermore, in a retrospective study done in China in 2021, maximum number (36.8%, 7/19) of LZD-resistant isolates with LZD MICs ranging from 4 to 8 mg/L had Cys154Arg substitution within *rplC* gene.^[16]

Other mutations which were not targeted in our study were *tsnR* gene loss-of-function mutations (*tsnR* gene encodes the 23S rRNA methyltransferase in MTB),^[17] *fadD32* mutations (codes for a fatty acyl-AMP ligase involved in the production of mycolic acid, a key component of the cell wall),^[18] and overexpression of efflux pumps. A potential source of discordance between phenotypic and genotypic LZD resistance could be because other mechanisms of resistance were not covered by the sequencing targets.

In the present study, 27 strains were phenotypically resistant but did not show any mutation. Such phenotypic resistance in

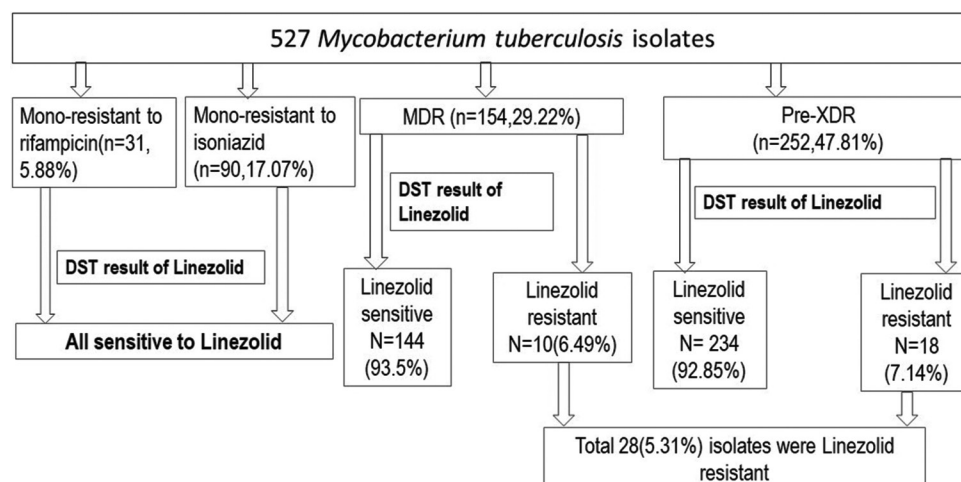


Figure 1: Distribution of linezolid resistance with respect to drug resistance category of *Mycobacterium tuberculosis* isolates

Table 1: Association of linezolid-resistance with clinical and microbiological outcome of patients at 6-month follow-up and with treatment regimen

	Linezolid resistant	Linezolid susceptible	Statistical analysis
Clinical outcome (n=56)	n=28	n=28	OR=1.00
Died	10 (35.71)	10 (35.71)	95% CI=0.34–2.98
Cured	18 (64.28)	18 (64.28)	Z statistic=0.00
			Significance level; P=1.00
Microbiological outcome (n=26)	n=12	n=14	OR=3.6667
Culture-positive	6 (50)	3 (21.42)	95% CI=0.6659–20.1911
Culture-converted	6 (50)	11 (78.57)	Z statistic=1.493
			Significance level; P=0.135
Treatment regimen (n=56)			
Short-oral regimen	14	23	χ^2 (DOF=1, n=56)=6.5225
Long-oral regimen	2	1	P=0.04
Change of regimen	12	4	

OR: Odds ratio, CI: Confidence interval, DOF: Degree of Freedom

the absence of genetic mutations may be due to the following reasons: (1) other resistance mechanisms such as efflux pump activity, (2) decreased permeability of the mycobacterial cell wall, (3) heteroresistance or mixed populations, (4) technical errors in pDST or in sequencing, (5) inaccurate reading by Nanopore method or, (6) presence of some unique mutations in the Indian isolates that could not be detected by the TB profiler pipeline.^[6]

The WHO TB mutation catalog 2021 offers guidance on single-nucleotide variations that are strongly predictive of resistance to anti-tuberculous drugs.^[19] Drug resistance leads to longer treatment durations and a higher risk of less favorable treatment outcomes in TB.^[20] As the number of resistance markers in a clinical isolate of MTB increases – reaching the threshold for XDR TB – patients face a significantly higher risk of treatment failure and death.^[21] Despite the substantial toxicity associated with LZD-based combination therapy, including risks such as myelosuppression and peripheral neuropathy, there are reports suggesting that this therapy can lead to better culture conversion rates and improved survival outcomes.^[22] The extensive daily medication regimen and prolonged duration of MDR-TB therapy elevate the risk of adverse drug reactions, primarily including ototoxicity and gastrointestinal issues, which can range from mild to severe.^[23] Given the slow growth of the MTBC and the high costs and complex infrastructure required for traditional pDST, rapid genotypic DST offers a more practical and efficient alternative for obtaining drug-susceptibility information essential for patient management.^[24]

In this study, we were able to get interim treatment outcomes in terms of mortality and culture conversion at the end of intensive phase of treatment for 26 TB cases. However, no association was found between clinical and microbiological outcomes and baseline LZD resistance. Change in drug regimen from shorter oral to longer oral was found to be significantly associated with primary LZD resistance. It is worth noting that even a child aged 9 years had a primary infection with LZD-resistant isolate. The development of LZD resistance without being exposed to it is a matter of concern.

It is suggested that unless the confidence levels of the sequencing method become high, only the pDST method should be used for guiding LZD treatment.

CONCLUSION

A moderate baseline resistance to LZD (5.3%) among the TB patients is alarming. Looking at the high baseline LZD resistance, it becomes imperative to conduct baseline pDST in all patients started on LZD therapy. Furthermore, agreement between pDST and genotypic nanopore sequencing resistance detection for LZD is low. LZD mutation in Indian isolates needs to be studied further and cataloged so that they can be included in the bioinformatics pipeline. Other mechanisms of LZD resistance should also be explored to possibly explain phenotypic LZD resistance.

Outcome of study

The study calculated low agreement between the phenotypically and genotypically detected LZD resistance. It also found that LZD resistance is not associated with clinical and microbiological outcomes of patients at 6-month follow-up.

Rationale of the study

LZD has gained an important position for the treatment of M/XDR TB. It becomes imperative to know the baseline resistance to LZD and the diagnostic modality for reliably detecting it.

Limitations of the study

The limitation of this study was that these 28 isolates could not be sequenced by the capillary sequencing method, which could have confirmed the presence of some unique mutations in the Indian isolates that have not yet been cataloged. Second, only the two most commonly reported LZD resistance genes could be assessed in the present study, and mutations in other genes such as fadD32 and tsnR and genes encoding efflux pump/transporters could not be assessed. Third, sequencing was done from isolates in a medium without LZD. Extracting DNA from resistant isolates grown in the LZD-containing

tube could have amplified minority variants resulting in better results, and finally, follow-up sample could not be obtained for 30 patients. The contact history or information on the living conditions of the LZD-resistant patients could not be obtained. Longer-term clinical outcomes and the impact of LZD resistance on treatment success could not be explained.

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Conflicts of interest

There are no conflicts of interest.

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Supplementary Table 1: The demographic details, drug resistance pattern, sequencing results, treatment regimen, treatment duration, and clinical and microbiological outcome of the patients

Lab ID	Age/sex	Drug resistance pattern	Sequencing result		Treatment regimen	Duration of treatment	Patient clinical outcome	Patient microbiological outcome
			rplC gene mutation	rrl gene mutation				
Phenotypic linezolid-resistant isolates								
148	23/male	MDR	WT*	WT*	Shorter-oral regimen	12 months	Died	NA
637	26/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	12 months	Cured	NA
801	30/female	Pre-XDR	WT*	WT*	Shorter-oral regimen to long oral regimen	Shorter oral (10 months) Longer oral (15 months)	Cured	Culture positive
1308	17/female	MDR	WT*	WT*	Longer-oral regimen	16 months	Cured	Bacteriological conversion
1378	21/female	MDR	WT*	WT*	Shorter-oral regimen	9 months	Cured	Bacteriological conversion
1515	66/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	11 months	Cured	NA
1520	51/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Cured	NA
1844	27/female	MDR	WT*	WT*	Shorter-oral regimen	11 months	Cured	NA
2075	41/male	Pre-XDR	Resistant (rplC p.Cys154Arg [0.93])	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (3 months) Longer oral (14 months)	Died	NA
2169	18/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Died	Culture positive
2680	17/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (8 months) Longer oral (15 months)	Cured	Bacteriological conversion
2911	32/male	MDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (9 months) Longer oral (19 months)	Cured	NA
2920	20/female	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (8 months) Longer oral (15 months)	Cured	Culture positive
3029	23/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (8 months) Longer oral (15 months)	Cured	Culture positive
3092	25/female	MDR	WT*	WT*	Shorter-oral regimen	11 months	Died	Culture positive
3293	12/female	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (10 months) Longer oral (17 months)	Cured	NA
4034	30/male	MDR	WT*	WT*	Shorter-oral regimen	10 months	Died	NA
4044	36/male	MDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (9 months) Longer oral (16 months)	Cured	Bacteriological conversion
4308	19/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (11 months) Longer oral (18 months)	Cured	NA
4377	16/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (9 months) Longer oral (16 months)	Cured	Bacteriological conversion
4380	10/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Died	NA
4499	23/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Died	NA
1819	45/male	Pre-XDR	WT*	WT*	Longer-oral regimen	17 months	Cured	Bacteriological conversion
2072	9/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (11 months) Longer oral (20 months)	Cured	NA
7306	50/male	MDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	NA
5474	25/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Died	Culture positive
2676	25/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	11 months	Died	NA
1907	35/male	MDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (11 months) Longer oral (18 months)	Died	NA
Phenotypic linezolid susceptible isolates								
1853	57/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	Bacteriological conversion
4308	20/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	Bacteriological conversion
5472	22/female	MDR	WT*	WT*	Shorter-oral regimen	5 months	Died	Bacteriological conversion
3823	19/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	NA
3222	27/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Cured	Bacteriological conversion
7376	32/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	11 months	Cured	NA

Contd...

Supplementary Table 1: Contd...

Lab ID	Age/sex	Drug resistance pattern	Sequencing result		Treatment regimen	Duration of treatment	Patient clinical outcome	Patient microbiological outcome
			rplC gene mutation	rrl gene mutation				
5760	20/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Died	Culture positive
2957	28/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	11 months	Died	Culture positive
1533	24/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (9 months) Longer oral (17 months)	Died	Culture positive
633	28/male	MDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (3 months) Longer oral (14 months)	Cured	NA
7131	50/male	MDR	WT*	WT*	Shorter-oral regimen	6 months	Died	Bacteriological conversion
2573	59/male	XDR	WT*	WT*	Shorter-oral regimen	8 months	Died	Bacteriological conversion
273	26/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	6 months	Died	NA
1210	27/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	5 months	Died	Bacteriological conversion
2520	47/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	Bacteriological conversion
2521	33/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	NA
420	36/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	3 months	Died	NA
647	23/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Cured	NA
1914	24/female	XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	Bacteriological conversion
5593	58/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	9 months	Cured	Bacteriological conversion
650	52/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	2 months	Died	NA
1694	36/male	MDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	Bacteriological conversion
2919	32/male	Pre-XDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (10 months) Longer oral (15 months)	Died	NA
3163	16/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	NA
3412	45/female	Pre-XDR	WT*	WT*	Longer-oral regimen	12 months	Cured	NA
3413	45/male	Pre-XDR	WT*	WT*	Shorter-oral regimen	10 months	Cured	NA
4215	32/male	MDR	WT*	WT*	Shorter-oral regimen to longer-oral regimen change	Shorter oral (10 months) Longer oral (16 months)	Cured	NA
4550	22/female	Pre-XDR	WT*	WT*	Shorter-oral regimen	3 months	Died	NA

*WT refers to the absence of any mutation in rplC and rrl genes. XDR: Extensively drug-resistant, MDR: Multidrug resistant, NA: Not available, WT: Wild type